

Project Requirement

September 23, 2025

1 Capstone Project: Data Analysis and Visualization Dashboard

1.1 Project Overview

Create a comprehensive data analysis and visualization dashboard that demonstrates mastery of Python programming, data manipulation, statistical analysis, and machine learning concepts covered throughout the course.

1.2 Project Requirements

1.2.1 Dataset

Students will work with a **sales and customer data** dataset containing: - Customer demographics (age, gender, location, income) - Product information (category, price, brand) - Sales transactions (date, quantity, revenue) - Customer behavior metrics (purchase frequency, average order value)

Dataset Size: ~10,000 records with 15+ features **Format:** CSV file provided (`sales_data.csv`)

1.2.2 Core Requirements

1. Data Loading and Exploration (20 points)

- Load the dataset using pandas
- Display basic dataset information (shape, columns, data types)
- Identify and handle missing values appropriately
- Perform exploratory data analysis (EDA) with summary statistics
- Create initial data quality assessment

2. Data Cleaning and Preprocessing (25 points)

- Clean inconsistent data formats
- Handle outliers using appropriate statistical methods
- Create derived features (e.g., customer lifetime value, purchase recency)
- Encode categorical variables appropriately
- Normalize/scale numerical features where needed

3. Statistical Analysis (20 points)

- Perform descriptive statistics for key metrics
- Conduct correlation analysis between variables
- Implement hypothesis testing (e.g., t-tests, chi-square tests)

- Calculate confidence intervals for key business metrics
- Identify statistically significant patterns

4. Data Visualization (25 points)

- Create at least 8 different types of visualizations using matplotlib/seaborn
- Include: histograms, scatter plots, box plots, heatmaps, time series plots
- Design interactive visualizations (optional bonus)
- Ensure all plots are properly labeled and formatted
- Create a dashboard layout organizing multiple visualizations

5. Machine Learning Implementation (30 points)

- Implement at least 2 different ML algorithms:
 - Classification: Predict customer segment or purchase behavior
 - Regression: Predict sales revenue or customer value
- Split data into training/testing sets appropriately
- Evaluate model performance with multiple metrics
- Visualize model results and feature importance
- Discuss model limitations and business implications

6. Business Insights and Recommendations (20 points)

- Summarize key findings from analysis
- Provide actionable business recommendations
- Identify opportunities for revenue growth
- Suggest customer retention strategies
- Create executive summary of insights

1.2.3 Deliverables

Required Files:

1. **analysis_notebook.ipynb** - Main Jupyter notebook with complete analysis
2. **requirements.txt** - Python dependencies
3. **README.md** - Project documentation and setup instructions
4. **presentation.pdf** - 10-15 slide presentation of findings

Optional Bonus:

- Interactive dashboard using Streamlit or Dash (+10 points)
- Advanced ML techniques (ensemble methods, feature engineering) (+5 points)
- Automated report generation (+5 points)

1.3 Input/Output Specifications

1.3.1 Input:

- **sales_data.csv** - Main dataset file
- Project requirements document
- Technical hints document

1.3.2 Output:

- Comprehensive analysis notebook
- Visualizations (saved as PNG/PDF files)
- Trained ML models (pickle files)
- Business insights report
- Presentation slides

1.4 Correctness Criteria

1.4.1 Technical Correctness (60%):

- **Code Quality:** Clean, well-commented, modular code
- **Data Handling:** Proper data cleaning and preprocessing
- **Statistical Methods:** Correct application of statistical tests
- **ML Implementation:** Proper train/test split, appropriate algorithms
- **Visualization:** Clear, informative, properly formatted plots

1.4.2 Analysis Quality (25%):

- **Insight Depth:** Meaningful business insights derived from data
- **Statistical Rigor:** Appropriate use of statistical methods
- **Interpretation:** Clear explanation of results and implications
- **Recommendations:** Actionable business recommendations

1.4.3 Presentation (15%):

- **Documentation:** Clear README and code comments
- **Organization:** Logical flow of analysis
- **Communication:** Effective presentation of findings
- **Professional Quality:** Polished deliverables

1.5 Grading Rubric

1.5.1 Excellent (90-100 points):

- All requirements met with exceptional quality
- Advanced techniques implemented
- Deep business insights with clear recommendations
- Professional presentation and documentation
- Bonus features implemented

1.5.2 Good (80-89 points):

- All core requirements met
- Solid technical implementation
- Good business insights
- Clear presentation
- Minor areas for improvement

1.5.3 Satisfactory (70-79 points):

- Most requirements met
- Adequate technical implementation
- Basic business insights
- Acceptable presentation
- Some technical or analytical gaps

1.5.4 Needs Improvement (60-69 points):

- Several requirements not fully met
- Technical implementation has issues
- Limited business insights
- Presentation needs work
- Significant gaps in analysis

1.5.5 Unsatisfactory (<60 points):

- Major requirements missing
- Poor technical implementation
- No meaningful insights
- Inadequate presentation
- Fundamental understanding gaps

1.6 Timeline

- **Week 1:** Data exploration and cleaning
- **Week 2:** Statistical analysis and visualization
- **Week 3:** Machine learning implementation
- **Week 4:** Business insights and presentation preparation

1.7 Submission Guidelines

- Submit all files in a single ZIP archive
- Ensure notebook runs without errors
- Include all generated visualizations
- Provide clear setup instructions in README
- Presentation should be 10-15 minutes for live demo

1.8 Academic Integrity

- Original work required
- Proper citation of external resources
- Collaboration allowed for discussion, but individual submission required
- Plagiarism will result in project failure